

MACHINE LEARNING AND DEEP LEARNING

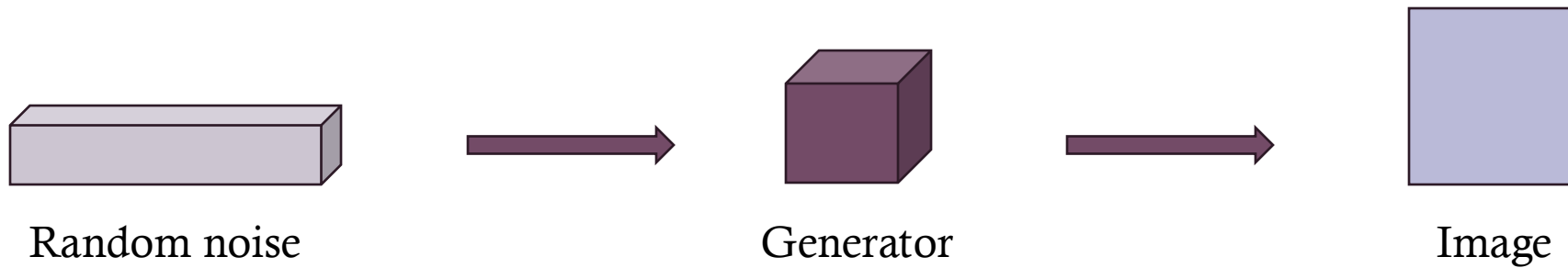
Lab 09: Deep Convolutional Generative Adversarial Network (DCGAN)

DCGAN

- As the name suggests, **Deep Convolutional Generative Adversarial Networks (DCGANs)** are generative models, able to generate new samples similar to those seen during training, but not exactly the same.
- This model relies on a **continuous conflict** between two sub-models: the **Generator** and the **Discriminator**.
- The role of the Generator is to **generate new, realistic samples**, similar to the ones present in the training set, while the Discriminator should learn to **distinguish** between real and fake images.
- If the Discriminator becomes too good, the Generator should improve as samples are probably not realistic enough.

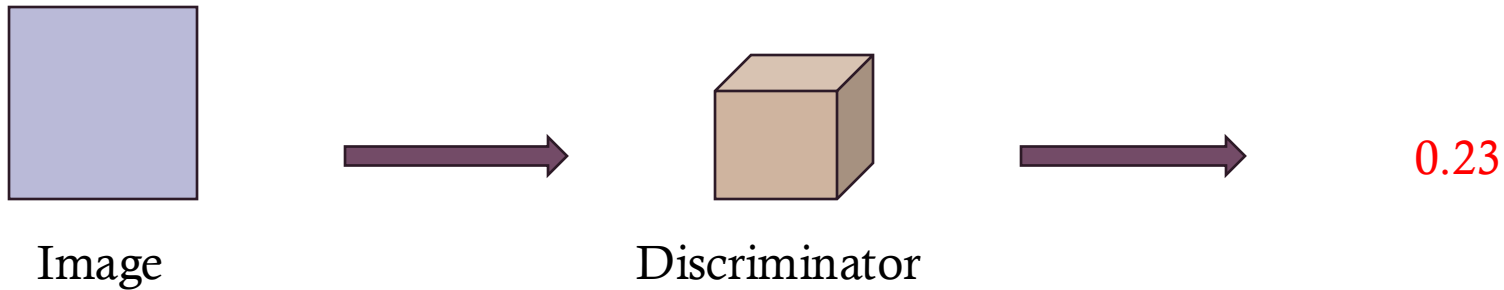
GENERATOR

- This model takes as input **compressed random noise** and decompresses it through Deconvolutions, expanding the spatial dimension and compressing the depth dimension, creating a new image.



DISCRIMINATOR

- The Discriminator takes as input both real and fake images and must be able to tell which ones are real and which ones are fake.
- For each image, it outputs a probability score between 0 and 1, where 0 means completely fake and 1 means completely real.



LOSS FUNCTION

- The employed loss is **Binary Cross Entropy (BCE)**.
- It follows the formula:

$$\max_G \mathbb{E}_{z \sim p_z(z)} [\log D(G(z))]$$

- Where:
 - Z is the random noise.
 - G(z) is the fake image generated by the Generator.
 - D(G(z)) is the belief of Discriminator over the generated fake image.

LOSS FUNCTION

- The final goal is to make the Generator able to construct fake images so realistic to **fool** the Discriminator.
- If this happens, $D(G(z))$ will be close to 1, therefore **the loss will be very small** (because $\log(1) = 0$). Otherwise, the loss will be high.
- **Note:** the Generator **never sees the training set**. When we say that it should generate images similar to the training ones it is because of the Discriminator's judgment: if generated images are different from training samples, the Discriminator will recognize them immediately as fake, and the loss will stay high.

TRAINING

- **Training** is usually performed over **two steps**. First, we **train the Discriminator** computing BCE loss over its predictions for both real and fake images, and we update its weights.
- Then, we train the Generator, using the loss function presented before.
- The **conflict** stays in the fact that we train the Generator to fool the Discriminator and at the same time we train the Discriminator not to be fooled by the Generator.

EXERCISE

- In this exercise you will construct a DCGAN and train it to generate samples from the EMNIST dataset (grayscale images of letters).
- **Task:** download *DCGAN.ipynb* and follow the instructions provided within the comments.